

# Case Study

## Title

### Cryptocurrency Price Tracker Using Quick Sort

## Aim

To analyze the application of Quick Sort in sorting cryptocurrency prices, study the impact of pivot selection on performance, and evaluate its efficiency using the Divide and Conquer technique.

## Procedure

1. Read the cryptocurrency prices.
2. Select a pivot element.
3. Partition the prices into three groups:
  - Less than the pivot
  - Equal to the pivot
  - Greater than the pivot
4. Recursively sort the left and right partitions.
5. Combine the sorted partitions.
6. Display the sorted prices and execution time.
7. Analyze the performance and optimization techniques.

## Algorithm

Step 1: Start.

Step 2: Read the list of cryptocurrency prices.

Step 3: Choose the middle element as the pivot.

Step 4: Divide the list into three partitions based on the pivot.

Step 5: Recursively apply Quick Sort to the left and right partitions.

Step 6: Merge the sorted partitions with the pivot.

Step 7: Display the sorted prices and execution time.

Step 8: Stop.

## Input

Cryptocurrency Prices:

45000 32000 58000 15000 27000 41000 39000

## Output

Sorted Prices:

[15000.0, 27000.0, 32000.0, 39000.0, 41000.0, 45000.0, 58000.0]

Execution Time: 0.08 ms

Average Time Complexity =  $\Theta(n \log n)$

Worst Case =  $\Theta(n^2)$

## Analysis

### Working of Quick Sort

**Quick Sort follows the Divide and Conquer strategy:**

- Select a pivot element.
- Divide the data into smaller partitions.
- Recursively sort each partition.
- Combine the sorted partitions to obtain the final sorted list.

### Issues in Cryptocurrency Price Tracking

- Cryptocurrency prices change rapidly, requiring frequent sorting.
- Poor pivot selection (such as always choosing the first or last element) can lead to highly unbalanced partitions.

- Unbalanced partitions increase execution time, especially for nearly sorted data.

## Performance

- Best Case:  $\Theta(n \log n)$
- Average Case:  $\Theta(n \log n)$
- Worst Case:  $\Theta(n^2)$  (when pivot selection is poor)

## Optimization Techniques

- Use a random pivot to reduce the chance of worst-case behavior.
- Use the median-of-three pivot (first, middle, last elements) for more balanced partitions.
- Switch to Insertion Sort for very small partitions to improve performance.
- Avoid unnecessary recursive calls by using optimized partitioning techniques.

## Result

Quick Sort efficiently sorts cryptocurrency prices using the Divide and Conquer approach. Its average time complexity of  $\Theta(n \log n)$  makes it suitable for handling thousands of dynamically changing cryptocurrency prices. Proper pivot selection techniques, such as random pivot or median-of-three, help avoid the worst-case complexity of  $\Theta(n^2)$  and improve overall performance.